

Rohit Raj

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Career Summary

Astrophysics researcher with 3+ years of experience in Python-driven data analysis, high-performance computing (HPC), and scalable data pipeline development. Proven ability to optimize workflows (50% faster processing), handle large datasets (220M+ stellar data points), and lead cross-functional teams. Skilled in Python, ML frameworks, and parallel computing, with a focus on translating complex problems into scalable software solutions.

Technical Skills

- **Programming:** Python (Astropy, SciPy, NumPy, Matplotlib, pandas), R, C++ (Intermediate)
- **HPC:** Parallel computing, job scheduling (Slurm), FDTD simulations (EM wave propagation)
- **Tools:** IRAF/DS9, Git, Mathematica, MATLAB
- **ML/AI (Intermediate):** PyTorch, scikit-learn, data modeling
- **Transferable:** Project management (NASA proposal), technical communication (AAS conferences), stakeholder alignment, abstract thinking & problem decomposition, logical reasoning & algorithmic thinking, data experimentation & analysis, rapid technical learning & adaptability, computational modeling & simulation

Professional Experience

University of California, Santa Cruz — Research Lead (Science Internship Program) Jun 2023–Present

- **Data Pipeline Optimization (2024):** Led spectroscopic and photometric analysis of 22M+ stellar data points from Keck/DEIMOS and Hubble, reducing processing time by 50% via Python automation.
- **Large-Scale Data Reconciliation (2023):** Developed systematic pipeline to compare and align velocity data from DESI and SPLASH surveys, enabling precise galactic merger history analysis.
- **Cross-Functional Leadership:** Mentored 11 high school/undergraduate interns across two summers, resulting in 4 conference presentations and 3 journal manuscript submissions.
- **Systems Integration:** Designed observational targets following calibration protocols for Keck adaptive optics and DEIMOS, significantly enhancing data accuracy for M33 star classification.

Imperial College London — Research Student (FDTD Simulations) Aug 2024–Present

- **HPC Modeling:** Developed EM wave propagation simulations for dynamic media, achieving 25% faster computation through parallel processing optimization.
- **Applied Physics:** Modeled quasi-crystal optical cloaking, providing insights for metamaterial R&D in aerospace.

NASA Proposal Team — Student Chief Scientist Sep 2024–Dec 2024

- **Technical Leadership:** Authored a technical mission proposal for NASA Marshall Space Flight Center, coordinating 12 researchers across 7 institutions. Participant in training sessions and proposal review panel.
- **Stakeholder Alignment:** Balanced engineering constraints with scientific goals, served on review panels.

Key Projects

Unusual Stars in M31/M32/M33 — Python, Spectral Analysis

- **Supervisor:** Professor Raja GuhaThakurta; Conducted spectral and photometric analysis of Wolf-Rayet & symbiotic stars using Hubble PHATTER (22M+ data points) and Keck DEIMOS; applied Gaussian PSF modeling to classify stellar winds and improve photometric accuracy.

DESI/SPLASH Survey Reconciliation — Python, HPC

- Supervisor: Professor Raja GuhaThakurta; Engineered a Python pipeline to reconcile discrepancies in DESI (fiber-based) and SPLASH (slit-based) stellar velocity data for M31, integrating Hubble PHAT/PHAST datasets (220M+ data points) to quantify measurement errors and refine galactic merger history models.

N-Body Simulation of Globular Clusters — MATLAB, Python

- Simulated million-year globular cluster evolution using N-body dynamics (HPC); visualized gravitational interactions and mass segregation effects via Matplotlib.

LIGO-Style Signal Detection — Python, Matched Filtering

- Supervisor: Professor Harsh Mathur; Reproduced LIGO's black hole merger detection using Python-based matched filtering (90% accuracy in noisy data); designed pedagogical framework for undergrad instruction.

Chaotic Double Pendulum Simulation — MATLAB

- Modeled nonlinear dynamics of a double pendulum in MATLAB; developed interactive module to demonstrate chaos theory.

Education

Juniata College — B.S. Physics — May 2025

- Study Abroad: University of Leeds (Theoretical Astrophysics)
- Relevant Coursework: Scientific Computing, Nonlinear Dynamics, N-body Simulations

Awards & Scholarships

- Herbert Levy Memorial Scholarship (*American Institute of Physics*), National Awardee (2024)
- Rufus Reber Physics Prize (*Juniata College*), Academic Excellence Awardee (2024)
- SPS-Google Scholarship (*American Institute of Physics*), Top 20 Nationwide (2023)
- Next Genius Scholarship (*India*), Selected from 4500+ applicants nationwide (2021)

Certifications & Training

- Machine Learning with Python: Foundations (LinkedIn, 2025); NASA Proposal Writing/Evaluation Training Program (2024); Career Essentials in Generative AI (Microsoft/LinkedIn, 2023); Canadian Astroparticle Physics Summer School (Queen's University, 2023)

Publications & Presentations

- First-author papers on Wolf-Rayet/symbiotic stars (*submitting to Astrophysical Journal*, Q1-Q3 2025)
- American Astronomical Society Conferences ('24/'25): Presented posters on galactic kinematics and spectral analysis

Outreach & Community Engagement

- The Business Show, London: Discussed astrophysics research and advocated for equity initiatives (Oct '24)
- Physics Phun Night: Organized science demos for 300+ attendees (Apr '22, Apr '23)
- U.S. Embassy New Delhi, India: As a US-India Dosti Ambassador, led STEM outreach for 1000+ students (Sep '23). Contributed to SPAN Magazine articles: "Looking Beyond Ivy Leagues" (Nov '22) and "Navigating Internships in the USA for International Students" (Nov '23)
- Shadow the Scientist, UCSC: Coordinated "Astronomy Night at W.M. Keck Observatory" outreach event in Kashmir, India, enhancing science engagement in conflict-affected area (Nov '22)
- Grassroots Emergency Response, India: Led team supporting 2,000+ migrant workers during pandemic; coordinated with government agencies. (May '20 - May '21)